EMPLOYEE ATTRITION PREDICTION

Big Data, Machine Learning, with Applications to Economics and Finance

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Data Description

The dataset we used in this study was taken from Kaggle. Its main aim is to support the development of predictive models for employee attrition, which is, in our opinion a critical human resource management challenge. The dataset was initially divided into separate training and testing subsets to facilitate model training and validation. The data structure is as following: there are 59 598 observations in the training set and 14 900 observations in the test set. Moreover, there are 21 initial variables and, in the training set 47.55% of the samples which have target variable “Attrition” equal to 1 (which means they left their jobs) and 52.45% of samples which stayed. Hence, there is no really class imbalance problem and there are neither explicit outliers nor missing values. As for the variables themselves, there are 4 continuous variables (“Age”, “Years at Company”, “Monthly Income” and “Distance from Home”) while the others are discrete and hence Ordinal and OHE encoders were applied to them.

EDA and feature engineering

From visual analysis we could see that “Age” and “Distance from Home” have almost symmetric distributions, “Years at Company” and “Monthly Income” are right-skewed and there are some minor outliers in the income variable, which have a pretty straightforward economic explanation, hence they were not dropped. As for the categorical variables, we could conclude that males are more prone to retain, while women have higher attrition. Secondly, people with ‘Fair’ work-life balance, “Low” job satisfaction and “Low” recognition are linked to higher turnover, while married employees are more likely to stay than single ones. What’s more, entry-level roles showed the highest attrition.

We were also able to observe some patterns in “Leadership Opportunities” and “Innovation Opportunities”. First of all, median monthly income between groups with and without such opportunities is the same which was confirmed by t-test (p-values are 0.27 and 0.52 respectively). However, there are far fewer people with these opportunities that without and most of them have entry job levels, which means that in our sample there is some kind of “Extra opportunities premium”, which defines the average income equality between these two groups. Moreover, we tested a very natural hypothesis that employees with remote work tend to stay using z-test. It showed us almost zero p-value for the hypothesis that proportion of people who left and worked remotely equals to the proportion of online workers who stayed against left-sided alternative. Thus, we can statistically-based assume that people with online work really appreciate this opportunity.

After completing EDA, we created new features, such as “Promotion rate”, “Years per Promotion”, “Stagnation Indicator”, “Income Adequacy Ratio”, “Zoomer” indicator, “Recognition Gap” and “Perfect Worker” indicator.

Modelling

First of all, to compare models with each other we use recall metric as it targets false negatives to be as small as possible which fully reflects our initial business goal. Then, we considered the first model - logistic regression, to account for linear relationships. The default logistic model was trained on the train set with all initial variables plus new variables and one principal component for variables connected to age (we will call it pca set). In order to prevent data leakage, we made a pipeline with StandardScaler and got recall score equal to 0.72 on 3-fold cross-validation. Using GridSearchCV we selected regularization hyperparameters and got in-sample recall equal to 0.74, while on inference – 0.7369. Hence, we can conclude that the logistic regression with the strong regularization (in our case C = 0.001 and penalty is ‘l1) results in good generalization power and quite good predictive power. Also, we additionally checked the accuracy score on inference which equaled 0.7369127 (the difference from recall is in the fifth digit), so this model is not ‘overfitted’ to the chosen metric. The most important variables appeared to be “Job Level” with a strong negative effect, “Remote Work_Yes” with also negative sign (as expected) and “Marital Status_Single” with positive sign (was also discussed above).

After that, we move to the first non-linear model – decision tree. Gridsearching for hyperparameters on the same set as before we got that the tree should use max_depth = 2, max_features = ‘log2’ and splitter = ‘random’, which is again a very strong regularization. On this parameter set we were able to achieve recall = 0.82 on inference, while accuracy was only 0.66. Even though we could achieve a high result, the model is definitely unstable due to the great implied stochasticity. That’s why, we considered the same grid search but without letting to choose the split type and left it ass default (‘best’). This model became much more complex with max_depth = 10, max_features = 0.5 and min_samples_split = 25. Nevertheless, the recall on inference dropped to 0.749, while accuracy increased to 0.736. For this model the most important feature turned out to be again “Job Level”, “Remote Work_Yes” and “Marital Statys_Single”. Also, “Work-Life Balance” and “Distance from Home” were quite important.

Ensemble Methods

After implementing simple models, we addressed more complex ones and decided to use RandomForest, XGBoost and CatBoost. What’s more, after that we used a self-made voting estimator for three sets of models – (logistic regression, decision tree, random forest), (logistic regression, decision tree, xgboost) and (logistic regression, decision tree, catboost) in order to account for linearity and non-linearity simultaneously.

Starting from random forest, we first select appropriate number of trees. In order to do that we evaluate recall score (and also accuracy just for completeness) on the tree’s parameter set (figure 1) using 3-fold cross-validation. As one can see, it worked as it should. Random forests generally do not hugely overfit, so the larger n_estimators the higher the metrics we use. The difference between n_estimators=325 and n_estimators=475/500 is not that big (on both recall and accuracy) so we will use the first in order to reduce the computational complexity

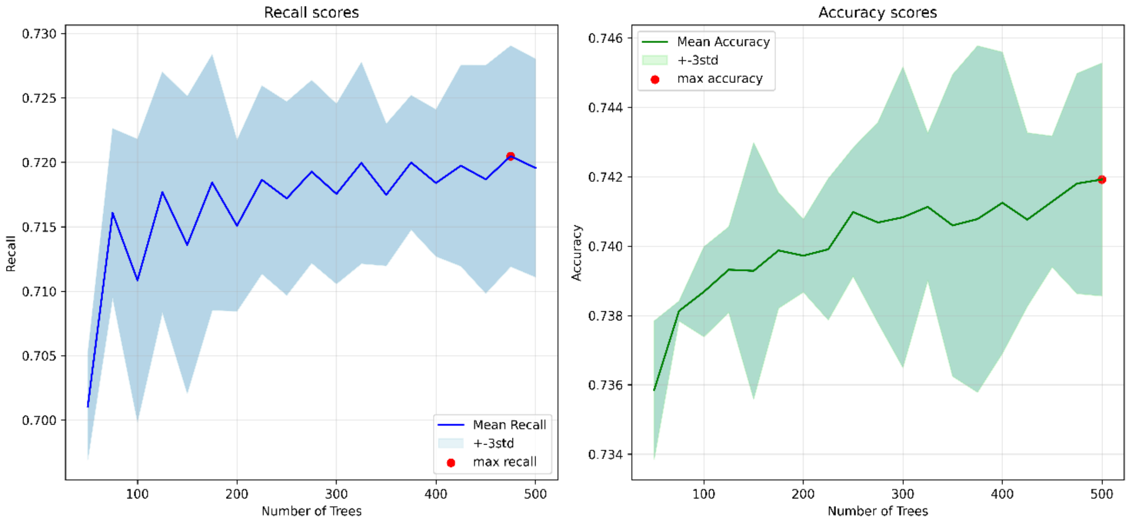

Figure 1. Number of trees optimization

After choosing number of trees to use, we moved to choosing the other hyperparameters. As now we use ensemble model, simple grid searching would work too long so we used optuna technique (which has Bayesian nature) on cross-validated recall score in order to select other hyperparameters. It showed that the biggest importance has max_depth hyperparameter, while the less important is max_features. Using optimized random forest, we again get “Job Level”, “Remote Work_Yes” and “Marital Status_Single” as very important features (job level has the biggest importance), while the least important are “Job Role”, “Stagnation”, both types of opportunities and “Employee Recognition”. Using the optimized model on inference we get a very interesting result, while both in-sample and out-of-sample accuracy increased (even though by a little margin – to 0.77 and 0.752), recall scores decreased to 0.746 and 0.722 respectively! So, in order to understand, whether there are just very simple relationships in our data or, vice versa, very complex and highly non-linear, we used boosting methods.

For XGBoosting we used optuna from the very beginning (figure 2) on cross-validated recall and after that used additional ‘against overfitting’ technique, early stopping, to get more robust model. Despite the powerful nature of the obtained model, recall on inference was still quite low for both recall 0.742 (though accuracy has increased to 0.756). As for CatBoost, which has built-in categorical encoder, we got out-of-sample recall equal to 0.741 and out-of-sample accuracy equal to 0.753. For both models similar features as before appeared to be important and not important at all. What’s more, the problem of low metrics was addressed through self-made voting estimators, which in turn resulted in almost the same results as above (almost 0.74 for out-of-sample recall and 0.75 for out-of-sample accuracy).

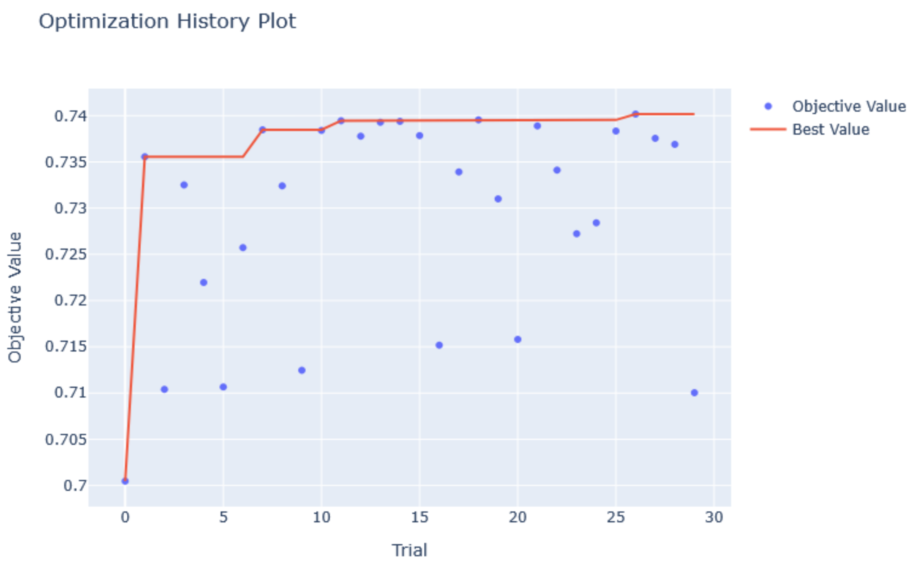

Figure 2. Optuna optimization

Validation

In order to assess the robustness of our ensemble models, we use 5-fold cross-validation to evaluate mean and standard deviation of our metrics. All three models (forest and boostings) appeared to be robust on both train and test sets. On the train set, random forest has mean recall equal to 0.72 (+- 0.006 std), xgboost - 0.74 (+- 0.005 std) and catboost – 0.742 (+- 0.007 std). On the test set the results for recall are the following: 0.71 (+- 0.017 std) for forest, 0.732 (+- 0.016 std) for xgboost and 0.733 (+- 0.013 std) for catboost. For accuracy scores the situation is similar.

Discussion

Firstly, as was discussed in the EDA part, having a remote job does affect the decision to stay. Also, being single makes people more mobile, while the higher job position, the smaller the incentive to leave arises. At the same time, some factors such as opportunities or recognition turned out to be almost unimportant for the decision-making.

Secondly, the most interesting thing in this project is that complex and highly non-linear models are not always the best ones. As one could see, a strongly regularized tree with some particular splits had more predictive power even than the CatBoost which is one of the most powerful classical ML models nowadays. Moreover, the other strong models like RandomForest and XGBoost had the same results. Hence, our dataset turned out to have some very niche patterns which are almost impossible to be modeled through complex structures. In fact, as we were able to see one simple or strongly regularized model can easily surpass ensemble models with many parameters. Nevertheless, all the models we built appeared to be very robust and what is more important, they all have very good generalization power as their results do not really differ between training and test sets.

To sum up, in this project we went through all the stages of machine learning methodology, starting from EDA and feature engineering and finishing with model tuning and validation. The most important insight we got – it’s not an axiom that complex and non-linear models can outperform basic (even though with regularization) linear or simple non-linear models. For the purpose of the project and its empirical implementation we would recommend to use a regularized decision tree or logistic regression since they are easily interpretable and perform well comparing to the black box models we had. Moreover, we strongly suggest that once an employer would like to use such models, human intelligence should not be forgotten as it may understand this very low-level patterns even better than the models we provided.